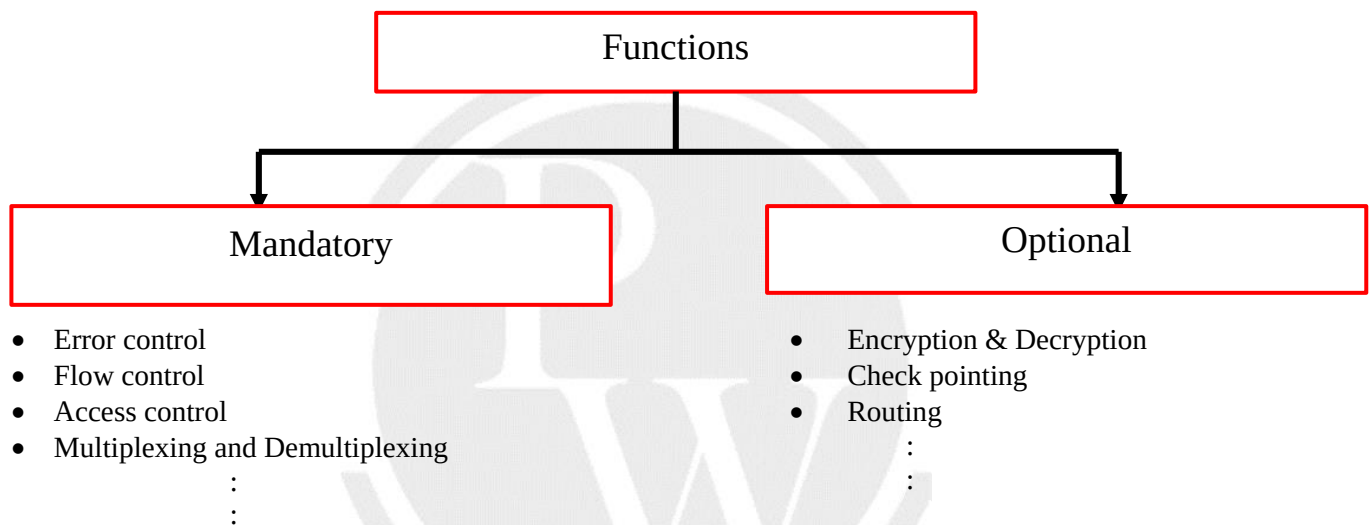


9

OSI AND TCP/IP PROTOCOL STACK

9.1 Functions of Computer Network



9.2 OSI/ISO

OSI : Open systems Interconnection model.

ISO : International Standards Organization. It is a multinational body dedicated to worldwide agreement on international standard.

9.2.1 OSI Mode

- This model has been proposed by ISO.
- An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture (Hardware/Software).
- The purpose is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware & software.
- This model has got 7 separate but related layers.

9.3 Functions of Physical layer

9.3.1 Physical Layer

Physical Layer is responsible for movement of individual bits from one Hop to next Hop.

9.3.2 Functions of physical Layer

1. It is used to define electrical, mechanical, functional and procedural characteristic of physical link.
(physical Link)
Copper → Electrical signal
Fiber → Light signal
Wireless comm. → Electromagnetic signal.
2. It defines transmission mode:
 - a. Simplex
 - b. Half duplex
 - c. Full duplex
3. It defines topology configuration:
 - Bus topology
 - Star topology
 - Mesh Topology
 - Tree Topology
4. It is totally Hardware layer.
5. It defines link configuration:
 - i Point to Point Link
 - ii Broadcast Link
6. It defines Encoding.
7. Bits Synchronization.
8. Bit rate control.

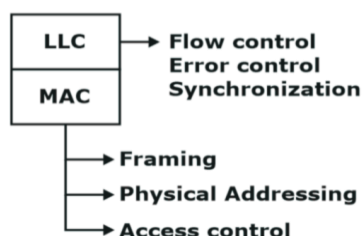
9.4 Data Link Layer

Data link layer is responsible for moving frames From One Hop (Node) to Next Hop (Node).

9.4.1 Function of data link layer

1. Flow control
2. Error control
3. Access control
4. Framing
5. Physical Addressing

Data Link Layer is divided into two parts





9.5 Network Layer

The network layer is responsible for the delivery of individual packet from source to destination.

9.5.1 Function of Network Layer

1. Host to Host connectively
2. Logical Addressing
3. Switching
4. Routing
5. Fragmentation
6. Congestion control:

9.6 Transport Layer

Transport Layer is responsible for process to process delivery. A process is an application program running on a host.

9.6.1 Function of Transport Layer

1. End to end connectively
2. Service point Addressing
3. Flow control
4. Error control
5. Segmentation and Reassembly
6. Congestion control
7. Connection Control
8. Multiplexing and Demultiplexing.

9.7 Session Layer

Session layer also known as **network dialog controller**. It establishes, maintains, synchronizes and terminates the interaction b/w sender and receiver.

9.7.1 Function of Session Layer

1. Authentication & Authorization
2. Check point or synchronization
3. Dialog control

9.8 Presentation Layer

This layer take care of syntax and semantics of the information exchange in between two communicating systems.

9.8.1 Function of Presentation Layer

1. Character translation
2. Encryption/Decryption
3. Compression



9.9 Application Layer

Application Layer is responsible for providing services to users. Users such as:

1. Mail services
2. File sharing
3. File transfer and many more

9.10 TCP/IP Model

This mode has got 5 different layer

Physical : No specific protocol

Data Link : No specific protocol

Network : ARP, RARP, ICMP, IGMP

Transport : TCP, UDP, SCTP

Application : DNS, SMTP, HTTP, FTP, POP, IMAP, SNMP, Telnet etc.

